

CubeTriage

Automated diagnosis for legacy OLAP cubes

— *in active development* —

One exported file in — a proposed star schema, a quantified complexity report and a human-review workbook out.
Every trap found, flagged and priced.

*Validated on Adventure Works 2019 Enterprise Edition —
120 review decisions found in ~2 seconds*

Malin Jacobsson

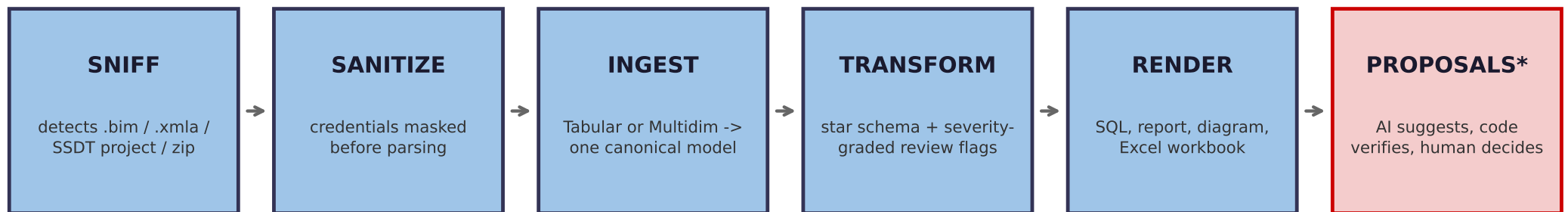
The problem

Migrating a legacy SSAS cube to a modern relational model costs a senior consultant weeks: untangling MDX, mapping role-playing dimensions, finding the KPIs and security roles nobody documented.

Before anyone can even price the migration, somebody spends days just answering: what is actually IN this cube?

CubeTriage answers that question in seconds — from a single exported metadata file. No cube data is read.

How it works



The first five stages are deterministic — same input, same output.
Proposals are opt-in behind a backend interface (mock / local LLM).

Outputs: *_canonical.json · *_out.json · *_schema.sql · *_load.sql ·
*_validation.sql · *_review.md · *_hitl.xlsx · *_diagram.png ·
*_decisions.json · *_run_report.json

Result on a real cube

Adventure Works 2019 Enterprise Edition (Multidimensional) —
Microsoft's own reference cube, built to demonstrate every
enterprise SSAS feature.

16 dimensions · 11 facts · 79 relationships · 38 measures

mapped automatically

22 calculated measures in MDX

flagged for manual translation — not ignored

29 role-playing · 4 parent-child · 4 reference · 6 m2m bridges

structural traps found

31 unmigratable constructs (KPIs, actions, perspectives)

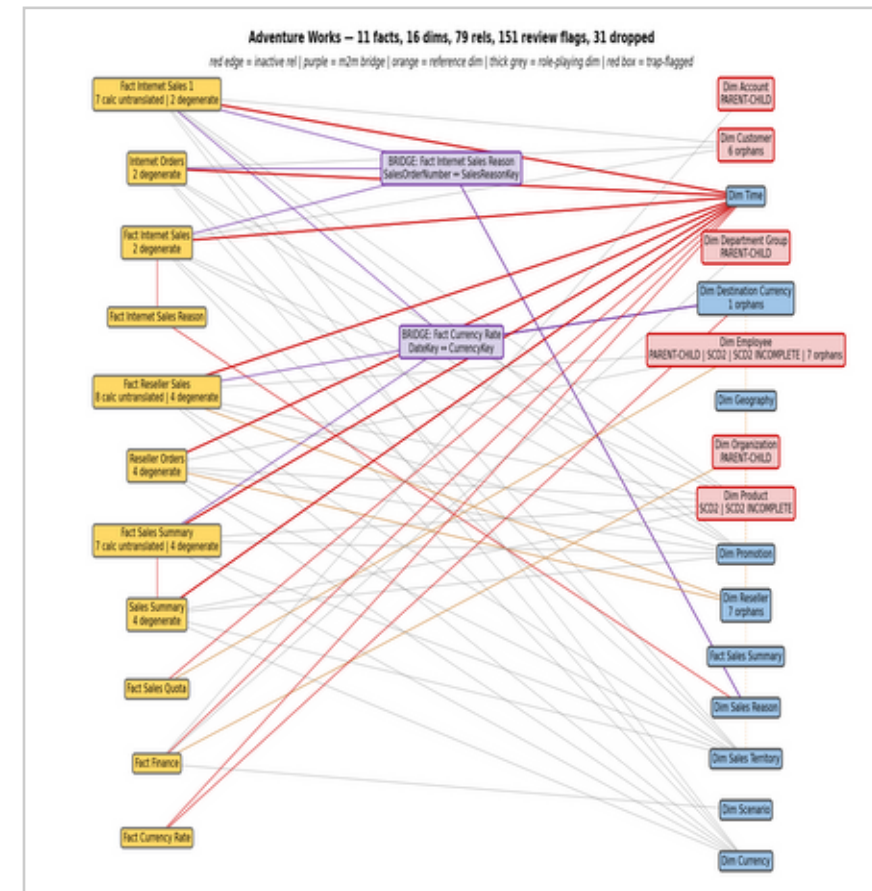
reported, never lost

101 inferred assumptions

each documented in its own channel

120 human-review items

the workbook IS the work order



Full pipeline run: ~2 seconds. Also verified on the
Tabular AdventureWorks .bim (compat 1400) — tables classified
purely from the relationship graph, no naming conventions needed.
Newest: generated load SQL + a 101-check validation harness.

The review loop

Every warn flag and dropped item becomes a numbered

decisions.json — stable ids, survives re-runs

Click a flag — see where the problem lives

canonical source slice + star-schema target table

Approve · reject · override — then regenerate

decided entries persist; unapplied overrides are reported, never written silently

Zero-dependency, offline, runs where the data lives

the Excel workbook stays as the exportable record

The screenshot displays a web application interface for managing data decisions. The top bar shows 'adventureworks' with '2 of 115 decided' and a 'Download decisions.json' button. Below this is a table with columns: 'open', 'approved', 'rejected', 'overridden', and 'all'. The 'open' column has a dropdown menu showing '113'. The table lists various decisions, each with a unique ID, a status, and a description. For example, 'DB7AE3F orphan_attribute' is listed with a status of 'open'. To the right of the table is a detailed view of the selected decision 'DB7AE3F - orphan_attribute'. This view shows the 'Dim Customer.Date First Purchase' dimension and a warning: 'Attribute not part of any hierarchy - keep or drop?'. It also displays the 'SOURCE' as 'MODEL/DIMENSIONS/DIM_CUSTOMER/ATTRIBUTES/DATE_FIRST_PURCHASE' and a JSON snippet:

```
{  "name": "Date First Purchase",  "source_column": "SimpleDateFirstPurchase",  "source_table": "dbo_DimCustomer",  "hidden": true,  "orphan": true}
```

. The 'TARGET' is 'dbo_DimCustomer.Date First Purchase'. The 'DECISION - CURRENT: OPEN' status is shown. At the bottom of the detailed view are buttons for 'Approve', 'Reject', and 'Override', along with a 'Comment...' field and 'Save comment' and 'Reopen' buttons.

Regeneration with decisions applied: ~1 second.

Design principles

Nothing is hidden

Constructs the engine sees but cannot migrate are recorded in a dropped channel. A reviewer always sees the full picture.

Provenance is structural

Observed facts, inferred values, dropped constructs and AI proposals are four separate channels — what the engine KNOWS vs GUESSES.

AI proposes — never decides

An optional backend may suggest resolutions. Code verifiers check every proposal against the model; the human approves.

Runs where the data lives

Deterministic, offline-capable, no telemetry — designed to run as a container inside the customer's own perimeter.